

1 - Creating Database

-- 1. Create and select database

```
CREATE DATABASE IF NOT EXISTS hospital_db;  
USE hospital_db;
```

-- 2. Drop dependent child table first to avoid Foreign Key Error 3730

```
DROP TABLE IF EXISTS Appointments;  
DROP TABLE IF EXISTS Doctors;
```

-- 3. Create Doctors table

```
CREATE TABLE Doctors (  
    doctor_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    department VARCHAR(50),  
    base_salary DECIMAL(10, 2),  
    bonus DECIMAL(10, 2),  
    allowance DECIMAL(10, 2),  
    consultation_fee DECIMAL(10, 2),  
    shift_hours INT  
);
```

-- 4. Insert sample data featuring Sanjana, Sahasra, and Ninada

```
INSERT INTO Doctors (doctor_id, name, department, base_salary, bonus, allowance,  
consultation_fee, shift_hours) VALUES  
(101, 'Sanjana', 'Cardiology', 90000.00, 5000.00, 2000.00, 500.00, 8),  
(102, 'Sahasra', 'Neurology', 110000.00, 8000.00, 3000.00, 600.00, 10),  
(103, 'Ninada', 'Pediatrics', 80000.00, 4000.00, 1500.00, 400.00, 6);
```

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```
SELECT
```

```
    doctor_id,  
    name,  
    base_salary,  
    bonus,  
    allowance,
```

-- Addition: Calculating Total Compensation

```
(base_salary + bonus + allowance) AS total_compensation,
```

-- Subtraction: Estimating Net Compensation after 5000 Tax Deduction

```
((base_salary + bonus + allowance) - 5000.00) AS net_compensation
```

```
FROM Doctors;
```

Result Grid							
		Filter Rows:			Export:	Wrap Cell Content:	
	doctor_id	name	base_salary	bonus	allowance	total_compensation	net_compensation
▶	101	Sanjana	90000.00	5000.00	2000.00	97000.00	92000.00
	102	Sahasra	110000.00	8000.00	3000.00	121000.00	116000.00
	103	Ninada	80000.00	4000.00	1500.00	85500.00	80500.00

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SELECT

name,

base_salary,

shift_hours,

-- Multiplication: Annual Base Salary

(base_salary * 12) AS annual_base_salary,

-- Division: Estimated Hourly Pay Rate (assuming 160 hours per month)

(base_salary / 160) AS estimated_hourly_rate,

-- Modulo: Checking if shift hours are odd or even (Remainder after division by 2)

(shift_hours % 2) AS shift_hour_remainder

FROM Doctors;

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content:
	name	base_salary	shift_hours	annual_base_salary	estimated_hourly_rate	shift_hour_remainder
▶	Sanjana	90000.00	8	1080000.00	562.500000	0
	Sahasra	110000.00	10	1320000.00	687.500000	0
	Ninada	80000.00	6	960000.00	500.000000	0

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-- Query 1: Equal (=) and Greater Than (>)

SELECT doctor_id, name, department, base_salary

FROM Doctors

WHERE department = 'Cardiology' OR base_salary > 85000.00;

-- Query 2: Not Equal (<>) and Less Than (<)

SELECT doctor_id, name, department, consultation_fee

FROM Doctors

WHERE department <> 'Pediatrics' AND consultation_fee < 600.00;

Result Grid Filter Rows: Edit:				
	doctor_id	name	department	consultation_fee
▶	101	Sanjana	Cardiology	500.00
*	NULL	NULL	NULL	NULL

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```
-- Query 1: Greater Than or Equal (>=) and Less Than or Equal (<=)
SELECT name, base_salary
FROM Doctors
WHERE base_salary >= 80000.00 AND base_salary <= 100000.00;
```

```
-- Query 2: Range matching (BETWEEN ... AND ...)
SELECT name, consultation_fee
FROM Doctors
WHERE consultation_fee BETWEEN 400.00 AND 600.00;
```

```
-- Query 3: List matching (IN)
SELECT name, department
FROM Doctors
WHERE name IN ('Sanjana', 'Ninada');
```

Result Grid Filter Rows:		
	name	department
▶	Sanjana	Cardiology
	Ninada	Pediatrics

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```
-- 1. Logical AND: Must satisfy ALL conditions
SELECT name, department, base_salary
FROM Doctors
WHERE base_salary > 85000.00 AND shift_hours >= 8;
```

```
-- 2. Logical OR: Must satisfy AT LEAST ONE condition
SELECT name, department, consultation_fee
FROM Doctors
WHERE department = 'Neurology' OR consultation_fee = 400.00;
```

```
-- 3. Logical NOT: Reverses the condition evaluation
```

```

SELECT name, department
FROM Doctors
WHERE NOT (department = 'Pediatrics');

```

Result Grid		Filter Rows:
	name	department
▶	Sanjana	Cardiology
	Sahasra	Neurology

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```

SELECT
    name,
    department,
    (base_salary + bonus) AS salary_plus_bonus,
    shift_hours
FROM Doctors
WHERE
    -- Comparison using an Arithmetic Expression inside Logical AND
    ((base_salary + bonus) > 90000.00)
    AND
    -- Logical OR condition combining shift hours and name selection
    (shift_hours > 7 OR name NOT IN ('Ninada'));

```

Result Grid

	name	department	salary_plus_bonus	shift_hours
▶	Sanjana	Cardiology	95000.00	8
	Sahasra	Neurology	118000.00	10